

第九章 醇、酚和醚

Alcohols, Phenols and Ethers

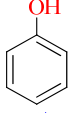


醇的分类、结构和命名

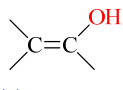
醇是羟基连接在**饱和 sp^3 碳原子**上的化合物



醇



酚



烯醇（不稳定）
（在醛酮中讲解）

• 醇的官能团是羟基(hydroxyl group) **$-OH$**

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醇的分类

3种分类方法：

1) 根据官能团所连烃基类型

RCH_2-OH	R_2CH-OH	R_3C-OH
伯醇	仲醇	叔醇
一级醇(1°)	二级醇(2°)	三级醇(3°)

3

醇的分类

2) 根据烃基结构

饱和醇	不饱和醇	芳香醇
$CH_3CH_2CH_2OH$	$CH_2=CHCH_2OH$	 $-CH_2OH$

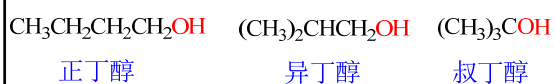
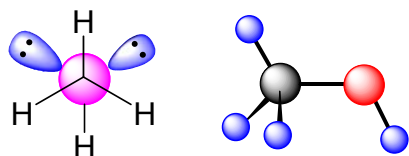
3) 根据羟基数目

一元醇	二元醇	多元醇
$R-OH$	$\begin{array}{c} CH_2CH_2 \\ \quad \\ OH \quad OH \end{array}$	$\begin{array}{c} CH_2CHCH_2 \\ \quad \quad \\ OH \quad OH \quad OH \end{array}$

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醇的结构

C、O都是 sp^3 杂化



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醇的物理性质

• 醇的沸点

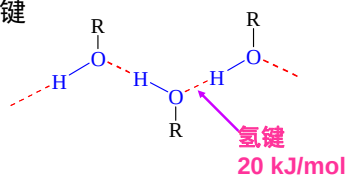
比同碳原子数的烷烃、卤代烃高，也比分子量相近的烷烃高

化合物	分子量	沸点(°C)
$\text{CH}_3\text{CH}_2\text{OH}$	46	78.5
CH_3CH_3	30	-98
$\text{CH}_3\text{CH}_2\text{CH}_3$	44	-42
$\text{CH}_3\text{CH}_2\text{Cl}$	64.5	12

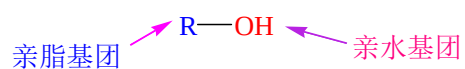
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醇的物理性质

• 醇的分子间氢键

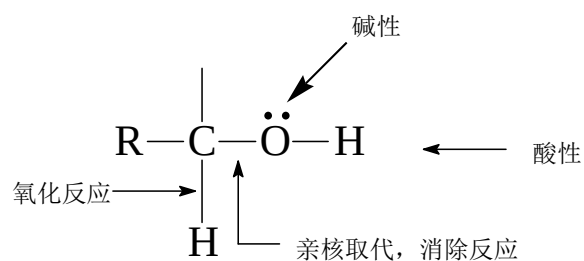


• 溶解度：低级醇溶于水，随分子量增加溶解度降低

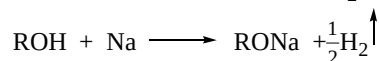
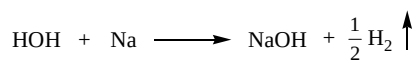


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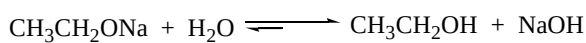
醇的化学性质



醇的酸性



反应速度 $\text{CH}_3\text{OH} > \text{CH}_3\text{CH}_2\text{OH} > \text{CH}_3\text{CH}_2\text{CH}_2\text{OH} > (\text{CH}_3)_2\text{CHOH} > (\text{CH}_3)_3\text{COH}$



醇是一种弱酸

普通醇与金属Mg反应（碘催化）可以制备无水的乙醇

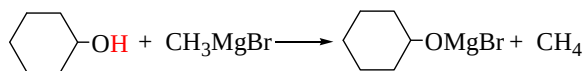
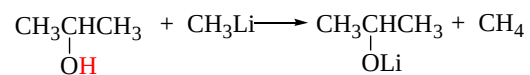
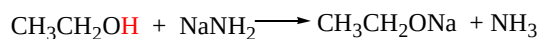
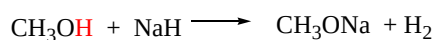
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酸性大小

化合物	pK _a	
(CH ₃) ₃ COH	18.00	弱酸
CH ₃ CH ₂ OH	16.00	
HOH	15.74	
CH ₃ OH	15.54	
CF ₃ CH ₂ OH	12.43	
(CF ₃) ₃ COH	5.4	
HCl	-7.00	强酸

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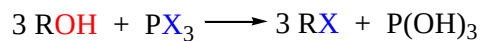
Example



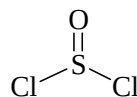
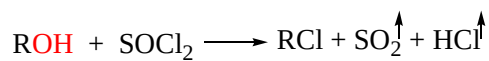
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取代反应

1. 与三卤化磷和氯化亚砷反应



或 PX_5 或 X_2 , P X = Br, I

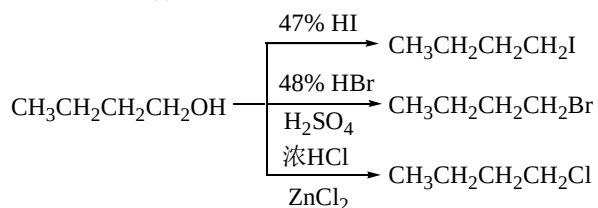


氯化亚砷
亚硫酸氯

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取代反应

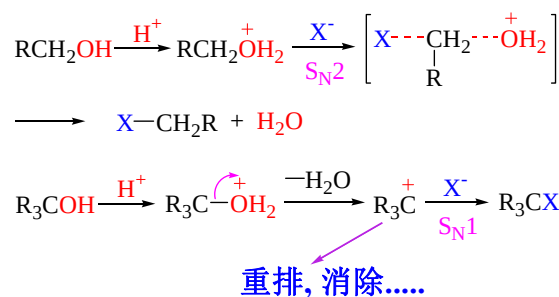
2. 与氢卤酸反应



反应活性: $\text{HI} > \text{HBr} > \text{HCl}$
 $3^\circ \text{ 醇} > 2^\circ \text{ 醇} > 1^\circ \text{ 醇}$

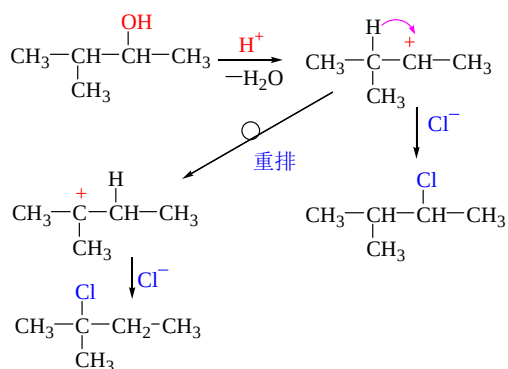
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反应机理



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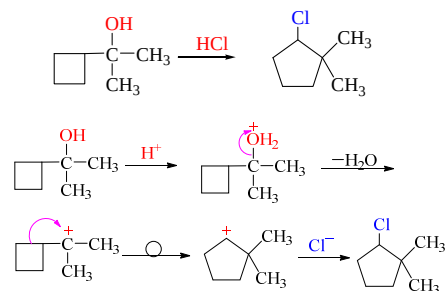
反应机理



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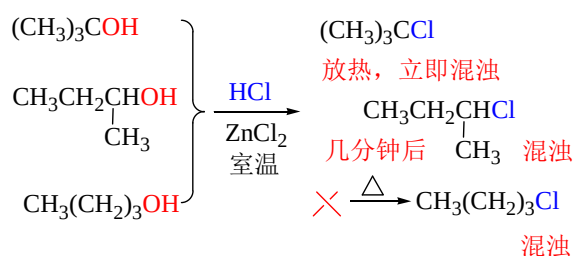
测试

请同学们用机理解释如下反应



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Lucas Reagent (浓HCl/ZnCl₂)



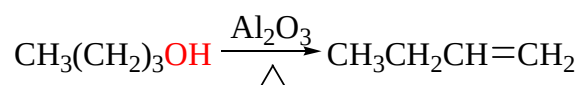
用于伯、仲、叔醇的鉴别
适用于C₄ ~ C₈的醇

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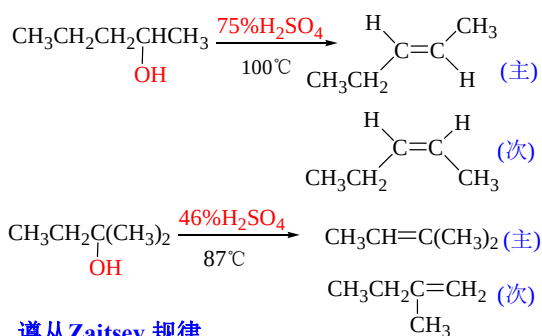
脱水反应

常用催化剂: H_2SO_4 , H_3PO_4 , Al_2O_3

1. 分子内脱水



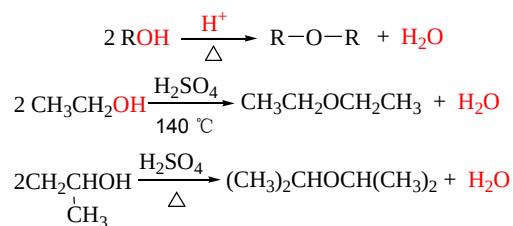
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脱水反应

2. 分子间脱水

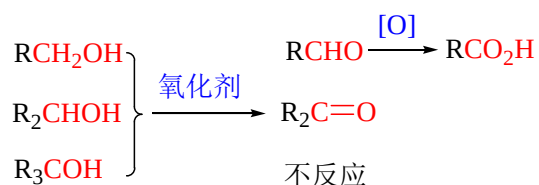


主要副反应：分子内脱水。
不适用于三级醇

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氧化与脱氢

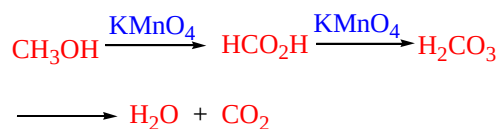
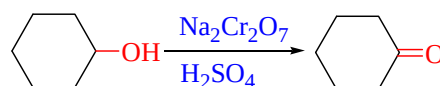
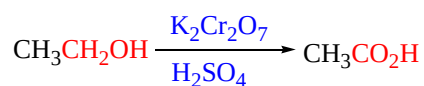
• 氧化剂氧化



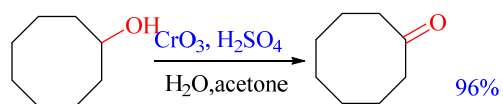
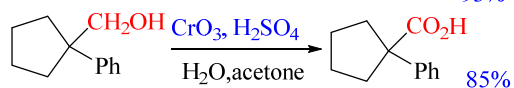
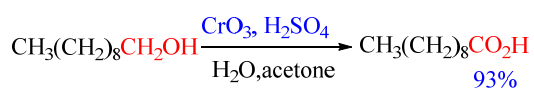
氧化剂: KMnO_4 , H_2CrO_4 , $\text{K}_2\text{Cr}_2\text{O}_7/\text{H}_2\text{SO}_4$
 $\text{Na}_2\text{Cr}_2\text{O}_7/\text{H}_2\text{SO}_4$, $\text{CrO}_3/\text{H}_2\text{SO}_4$

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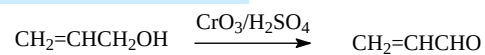
Example



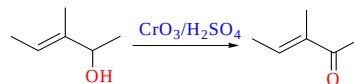
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Jones' reagent — $\text{CrO}_3/\text{H}_2\text{SO}_4$ 

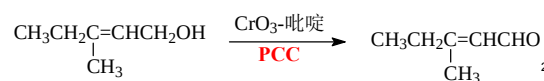
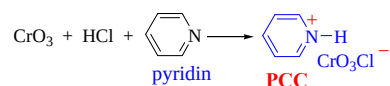
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Jones试剂: $\text{CrO}_3/\text{H}_2\text{SO}_4$ 

丙烯醛沸点较低, 可以蒸馏得到; 如不蒸馏, 得到丙烯酸。



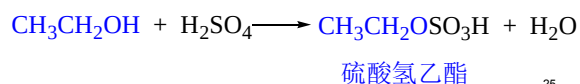
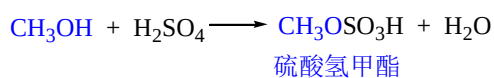
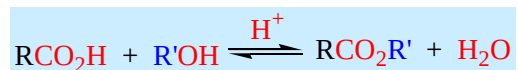
PCC (Pyridinium Chlorochromate)



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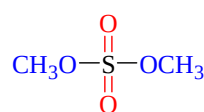
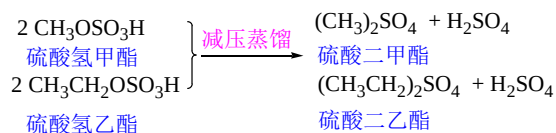
成酯反应

醇与含氧酸(有机酸、无机酸)反应成酯



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用途

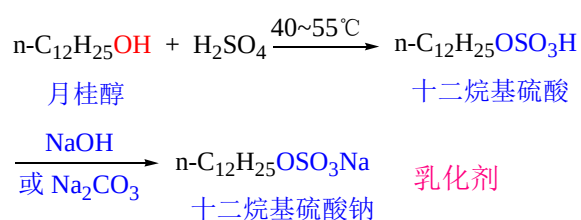


烷基化试剂

剧毒!

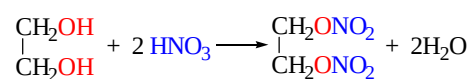
26

用途



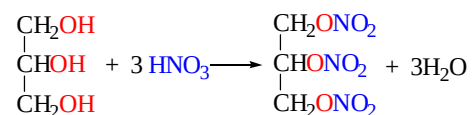
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用途



乙二醇

乙二醇二硝酸酯



甘油

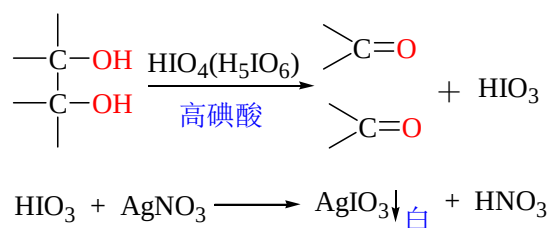
甘油三硝酸酯
硝化甘油

烈性炸药

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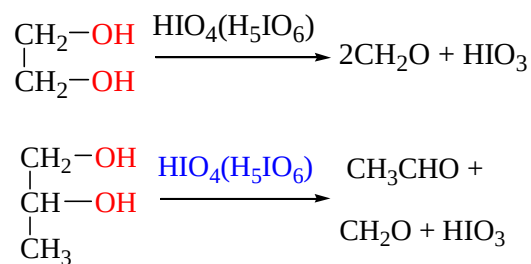
邻二醇的反应

1. 被高碘酸氧化



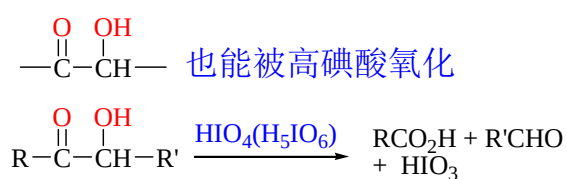
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Example



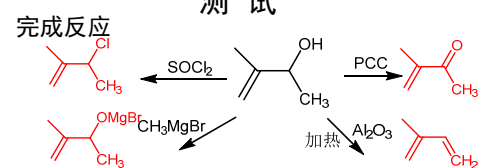
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Example

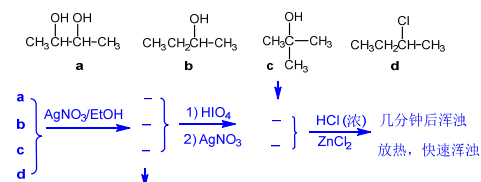


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测试



用化学方法鉴别如下化合物

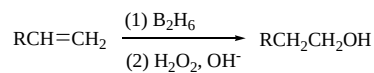
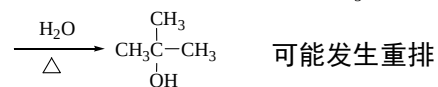
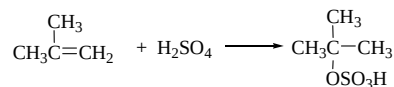
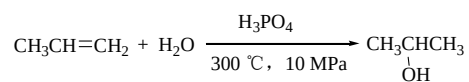


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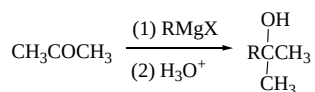
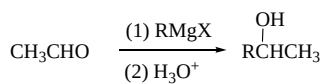
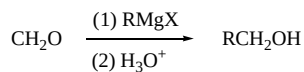
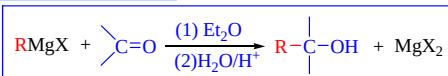
醇的制备和来源

- 由烯烃制备——水合，硼氢化-氧化
- 由羰基化合物制备——Grignard 反应，与炔化物反应，还原
- 由卤代烃水解

由烯烃制备

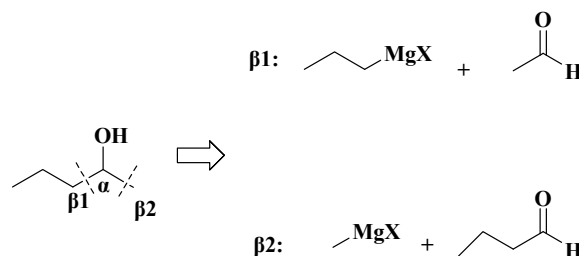


由羰基化合物制备



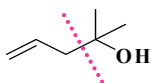
Example

逆向合成分析



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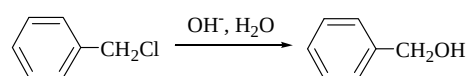
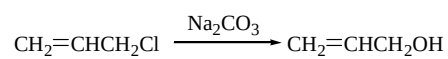
Example



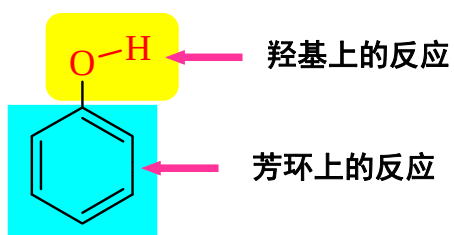
由卤代烃水解

- 副反应多
- 一般情况下醇比卤代烃易得

由卤代烃水解主要应用到如下类型的化合物中：

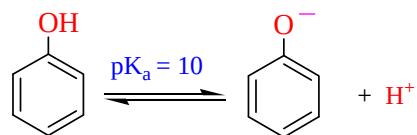


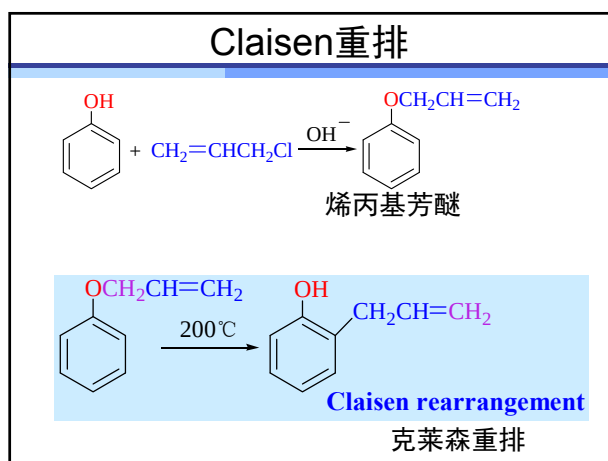
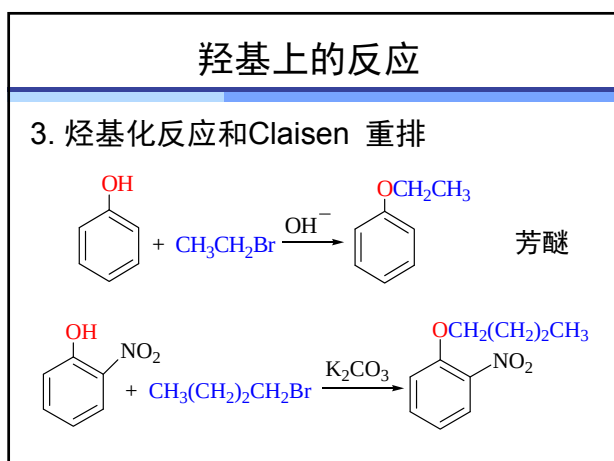
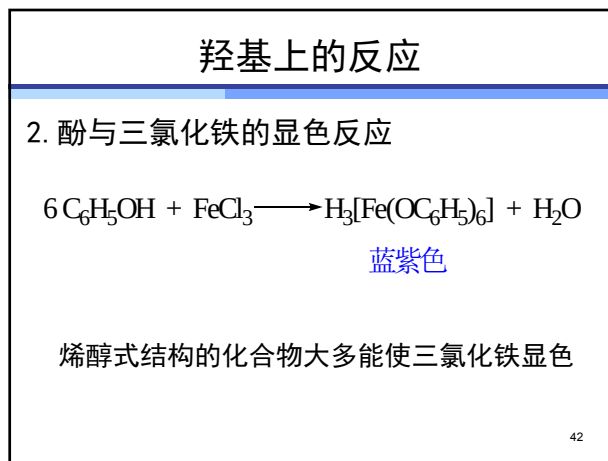
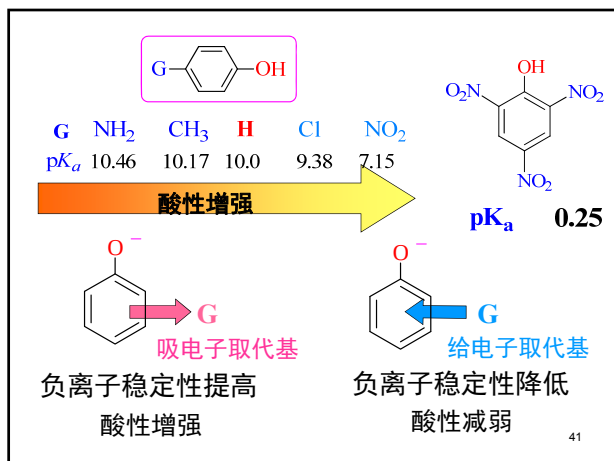
酚的化学性质



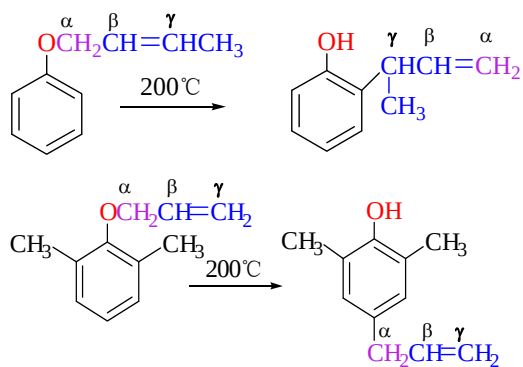
羟基上的反应

1. 酚的酸性

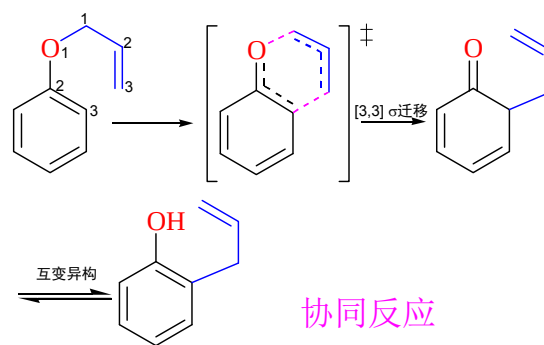




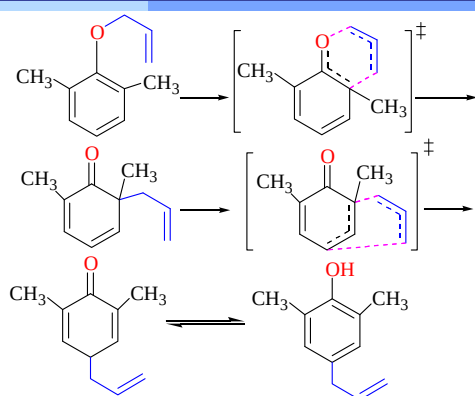
Example



反应机理



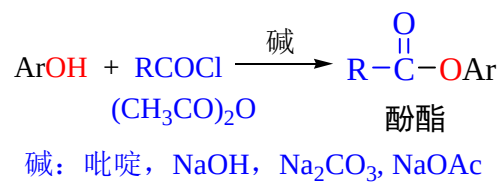
反应机理



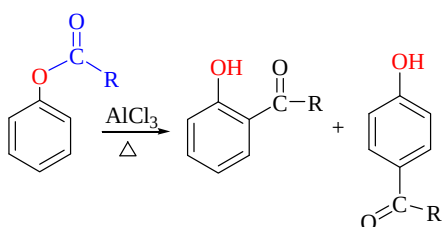
羟基上的反应

4. 酰基化反应及Fries重排

一般用酰氯或酸酐反应

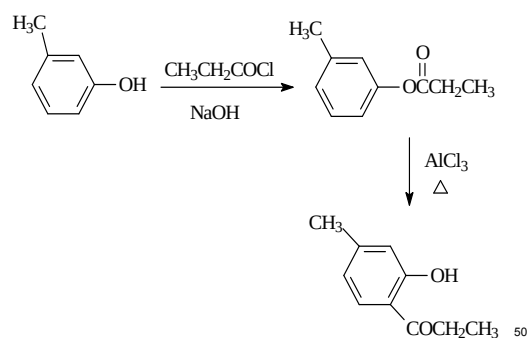


Fries重排



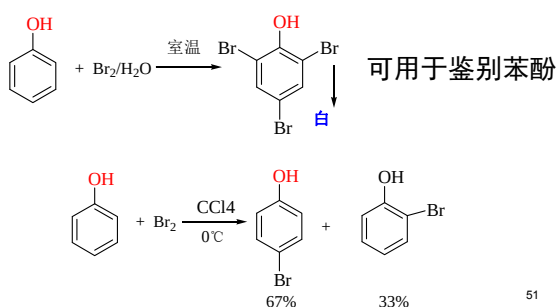
高温下产物以邻位为主
较低温度下产物以对位为主

Example



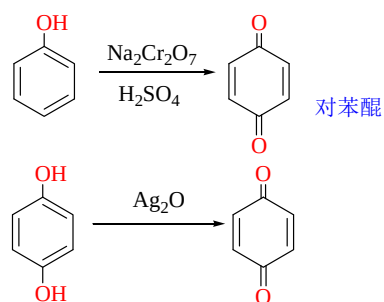
芳环上的取代反应(自学)

卤化反应



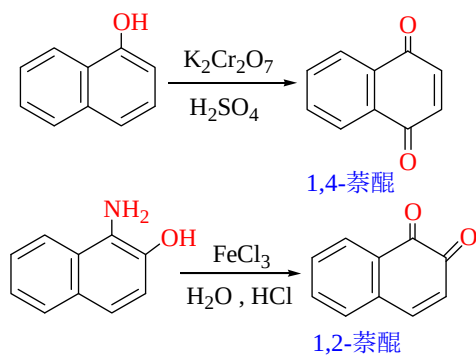
51

酚的氧化反应



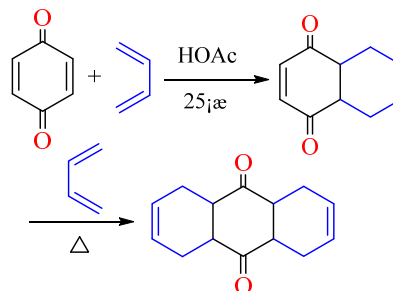
52

Example



53

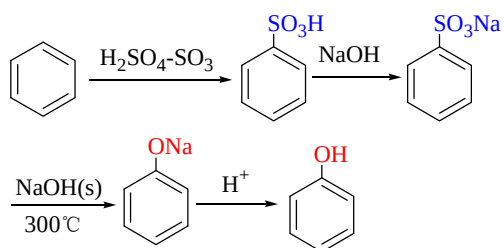
对苯醌可作为亲双烯体
发生Diels—Alder反应



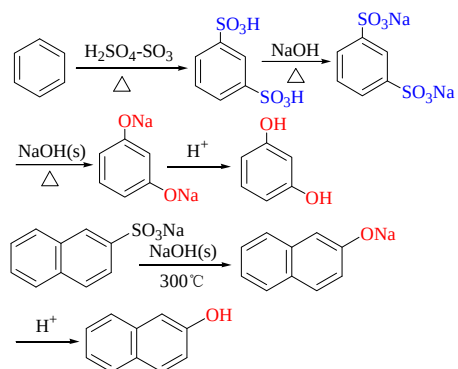
54

酚的制备

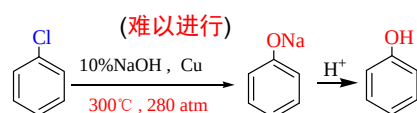
1. 磺酸盐碱熔法



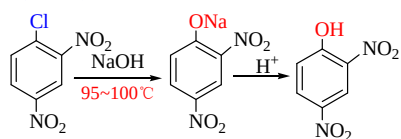
Example



2. 氯苯水解法

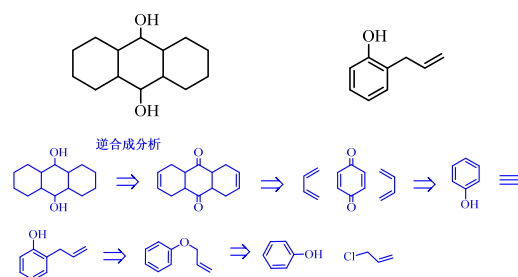


卤素的邻、对位有强吸电子基团时，
水解变得容易进行：



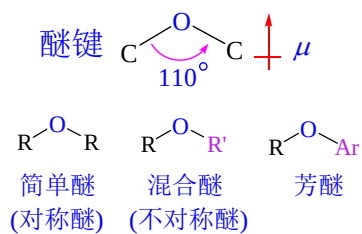
测试

以苯酚为原料，合成下列物质



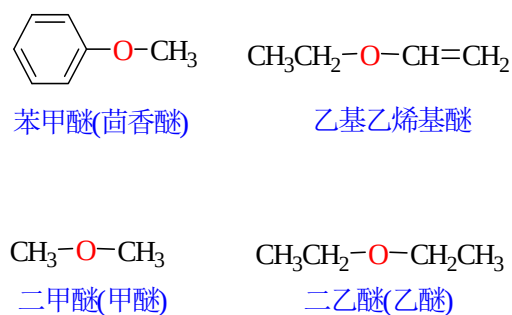
58

醚



59

命名（了解）



60

命名 (了解)

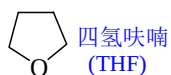
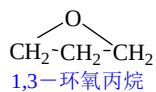
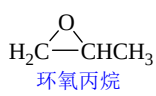
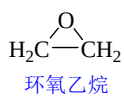
2. 环醚

环醚

环氧化合物

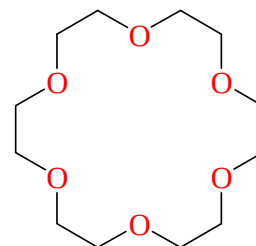
大环多醚

环氧化合物



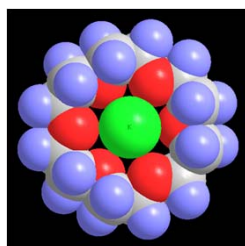
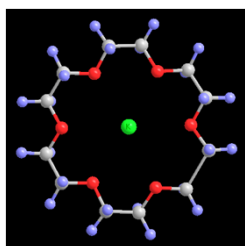
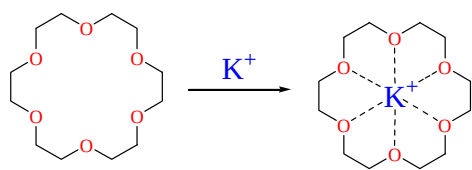
61

大环多醚



18-冠-6
18-crown-6

62



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醚的物理性质

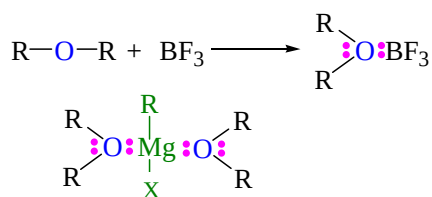
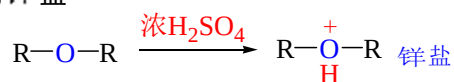
- b.p. 比同碳原子数的醇低得多
- 水中溶解度小(环醚除外)

	CH_3OCH_3	$\text{CH}_3\text{CH}_2\text{OCH}_2\text{CH}_3$
b.p.(°C)	-24.9	34.6
	$\text{CH}_3\text{CH}_2\text{OH}$	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$
b.p.(°C)	78.4	117.8

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醚的化学反应

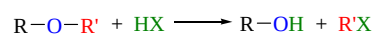
1. 形成铯盐



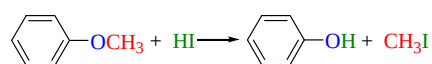
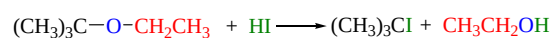
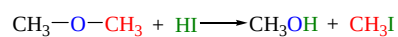
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醚的化学反应

2. 醚键的断裂



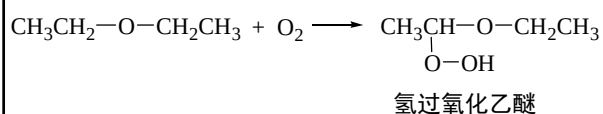
反应活性 $\text{HI} > \text{HBr} > \text{HCl}$



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醚的化学反应

3. 过氧化物peroxide的生成

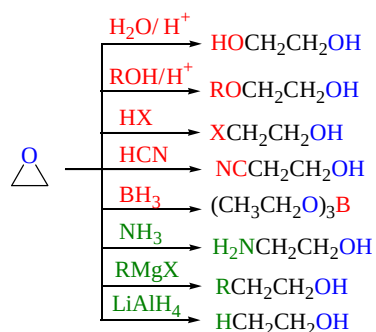


- 检验：淀粉—碘化钾试纸
- 除去过氧化物的方法：FeSO₄洗涤
- 储存时加Fe可防止形成过氧化物

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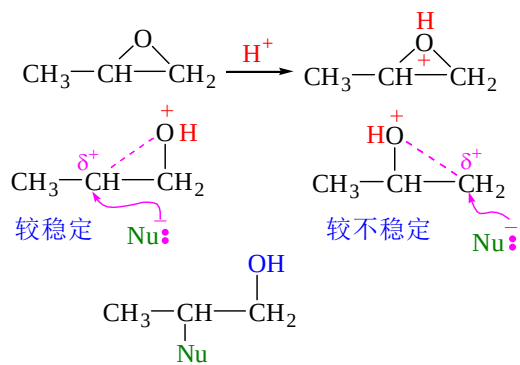
醚的化学反应

4. 环氧化合物的反应



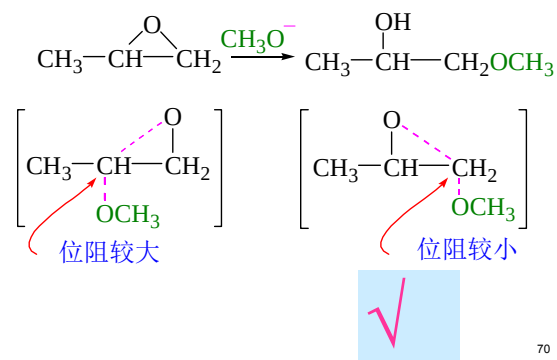
68

酸催化下的开环反应



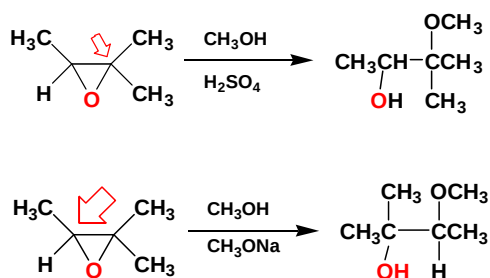
69

碱催化下的开环反应



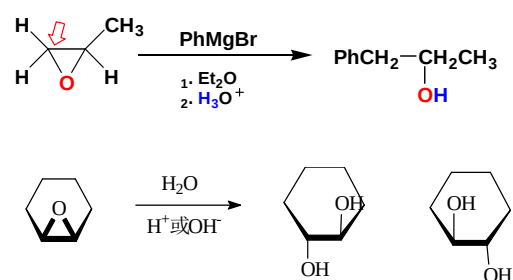
70

Example



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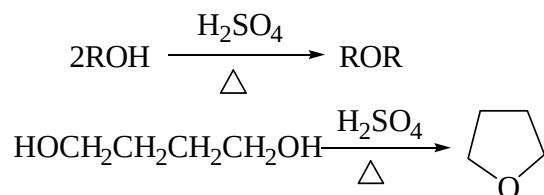
Example



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醚的制备

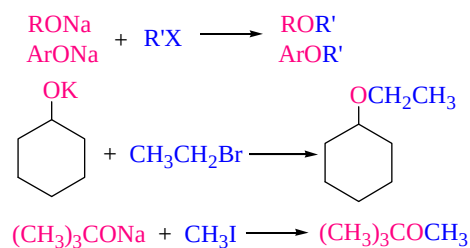
1. 醇的分子间脱水



73

醚的制备

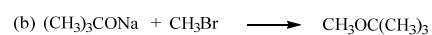
2. Williamson 合成法



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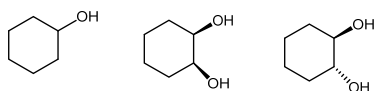
测试

1、下列两个制备甲基叔丁基醚的反应，哪个产率高？



b) 产率高; a) 消除为主

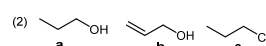
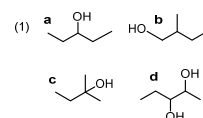
2、用环己烯制备如下化合物



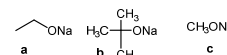
75

作业 (一)

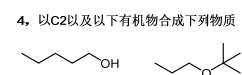
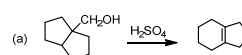
1. 用简单的化学方法鉴别下列各组化合物



2. 按碱性由强到弱排序



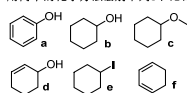
3. 写出下列反应机理



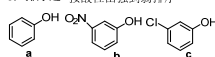
76

作业(二)

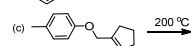
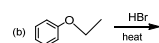
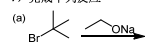
5. 用简单的化学方法鉴别下列6个化合物



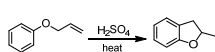
6. 排序题 按酸性由强到弱排序



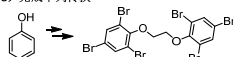
7. 完成下列反应



8. 写出下列反应机理



9. 完成下列转换



10. 以苯, C3及以下有机物合成下列物质

